The Siamese Cat Mystery

Name _____ Period ___

Focus question: How can a difference in DNA change which parts of a cat are dark?

Warm-up: Notice and wonder

Part 1: Data 1, temperature and fur color

tape the temperature table here

Part 2: Data 2, litters of kittens

tape the litter table here

Questions I still have

1. _____
2. _____

Part 3: Key ideas and my first model

Five things I learned today

1. _____
2. _____
3. _____
4. _____
5. _____

What Is DNA Made Of?

Name _____ Period ___

Focus question: What is DNA built from, and what is the message?

Warm-up: What does DNA look like?

Part 1: One nucleotide

Part 2: Base pairing, write the partner strands

Questions I still have

1. _____
2. _____

Part 3: One letter apart

Five things I learned today

1. _____
2. _____
3. _____
4. _____
5. _____

Build, Copy, and Read a Gene

Name _____ Period ___

Focus question: How does a gene's message become a chain of amino acids?

Warm-up: Copy, then read

Part 1: Base key and code table

tape the code table here

Part 2: Lab, build the gene, copy it, read it

Questions I still have

1. _____
2. _____

Part 3: Change one letter, then connect to the cat

Five things I learned today

1. _____
2. _____
3. _____
4. _____
5. _____

What Can One Letter Do?

Name _____ Period ___

Focus question: What can a one-letter change in DNA do?

Warm-up: What changed in Part D?

Part 1: Three things a one-letter change can do

Part 2: Card sort, cards A to F

glue the sorted cards here

Questions I still have

1. _____
2. _____

Part 3: The Siamese cat, one amino acid different

Five things I learned today

1. _____
2. _____
3. _____
4. _____
5. _____

Same DNA, Different Jobs

Name _____ Period ___

Focus question: If all cells have the same DNA, why do cells have different jobs?

Warm-up: Skin cells and nerve cells

Part 1: Two kinds of cells

tape the cell table here

Part 2: Cells in a Siamese cat's ear

Questions I still have

1. _____
2. _____

Part 3: Where does the changed protein work?

Five things I learned today

1. _____
2. _____
3. _____
4. _____
5. _____

Case Data

Name _____ Period ___

Focus question: How does a DNA change lead to an effect on the body?

Warm-up: The chain for the cat

Part 1: Case 1, Siamese and solid cats

tape the case table here

Part 2: Case 2, a human gene

Questions I still have

1. _____
2. _____

Part 3: Claim, evidence, reasoning

Five things I learned today

1. _____
2. _____
3. _____
4. _____
5. _____

Review and Back to the Cat

Name _____ Period ___

Focus question: How can I explain the Siamese cat with DNA, genes, and proteins?

Warm-up: My CFA scores and my reviews

Part 1: Vocabulary

Part 2: My revised model

Questions I still have

1. _____
2. _____

Part 3: Write It, two Siamese kittens

Five things I learned today

1. _____
2. _____
3. _____
4. _____
5. _____

Unit Assessment and Reflection

Name _____ Period ___

Focus question: What do I know about how DNA leads to traits?

Warm-up: Before the test

Part 1: Back to the cat, what I would add

Part 2: Looking back at my CFAs

Questions I still have

1. _____
2. _____

Part 3: What I will do next time

Five things I learned today

1. _____
2. _____
3. _____
4. _____
5. _____